

Business Problem Overview

A retail organization aims to gain deeper insights into customer purchasing behavior to enhance sales performance, strengthen customer relationships, and improve retention rates. The company has observed variations in buying habits across different customer groups, product categories, and purchasing channels, including both online and in-store transactions.

Management wants to understand the key factors influencing customer purchase decisions, such as promotional offers, product ratings, seasonal trends, and preferred payment methods. By identifying these patterns, the company hopes to improve customer engagement, optimize marketing campaigns, and refine product strategies.

Business Objective

"How can customer purchasing data be utilized to uncover valuable trends, enhance customer experiences, and support more effective marketing and product decisions?"

Project Deliverables

1. Data Cleaning and Transformation (Python)

- Prepare the dataset by handling missing values, duplicates, and inconsistencies.
- Transform the data into a suitable format for analysis.

2. Data Exploration and Analysis (SQL)

- Organize the data into structured tables.
- Perform queries to identify customer segments, purchasing behavior, and factors influencing sales.

3. Interactive Dashboard Development (Power BI)

- Create an interactive dashboard to visualize customer trends, sales performance, and key business metrics.
- Present insights through clear visualizations to support decision-making.

4. Project Presentation

- Prepare a presentation summarizing the project approach, key findings, business insights, and actionable recommendations for stakeholders.